



Architecture & Development Proposal

Enterprise Traffic Analytics Portal: A secure, multi-tenant web portal & Progressive Web App for regional traffic sensor analytics.

 PROJECT PROPOSAL

PROPOSAL HIGHLIGHTS

₹400,000

TOTAL COST (APPROX.
€4,450)

4 Phases

MILESTONE ROADMAP

€100 / mo

SLA & MAINTENANCE

PREPARED FOR



Data Ferns

PREPARED BY

Isotope 15 Services

DATE

June 11, 2026



1. Executive Summary

System definition and core business objectives

This proposal outlines the development of a secure, multi-tenant web portal and Progressive Web App (PWA) designed to ingest, process, and visualize regional traffic sensor data. The platform utilizes localized mapping, interactive dashboards, and strict role-based data isolation, allowing end-customers to securely access proprietary traffic analytics from any desktop or mobile device.



Multi-Tenant Isolation

CORE

Strict separation of customer accounts ensuring proprietary traffic log assets and regional reports remain private, secure, and uncompromised.



PWA Capabilities

DELIVERY

End-users can securely install the portal directly to their iOS or Android home screens without navigating Apple or Google App Stores.



Localized Mapping

INTERACTIVE

Bypasses commercial API licensing costs (Mapbox, Google Maps) entirely by utilizing high-velocity, zero-cost OpenStreetMap tiles.



Sensor Visuals

ANALYTICS

Ingests parsed data packets directly from the desktop application, generating dashboard charts, traffic flows, and speed indicators.



Development Philosophy

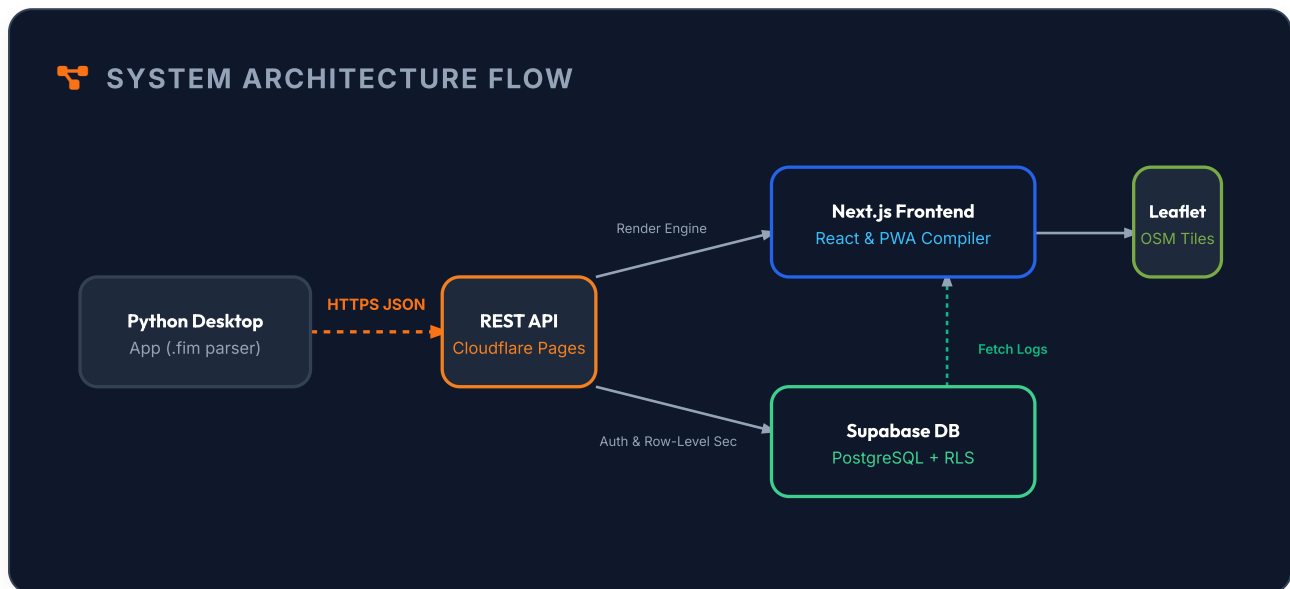
By coupling localized map tiles with serverless databases, we eliminate recurring license fees while keeping page loads instant and data access fully locked down.



2. Technical Architecture & Tech Stack

High-speed data rendering and cost mitigation roadmap

The technology stack is engineered specifically to eliminate recurring commercial mapping API costs, guarantee GDPR data sovereignty, and deliver high-velocity rendering for large datasets.



Next.js / React



Leaflet.js + OSM



Supabase (Postgres)



Cloudflare Pages

Cost-Saving Strategy: Leaflet.js paired with OpenStreetMap tiles completely bypasses commercial mapping providers. By serving map layers directly from open caches, we guarantee **zero billing impact** regardless of visitor count or traffic volume peaks.



3. GDPR Compliance & Data Sovereignty

Ensuring security and regulatory compliance for telemetry records

To guarantee compliance with European Union data protection regulations (GDPR) regarding the storage, processing, and handling of proprietary regional traffic sensor records:



EU Data Localization

COMPLIANT

All relational records, GPS coordinates, and user profile data will be provisioned exclusively within European hosting zones. We will deploy the primary database clusters and auth servers inside the **AWS/Supabase eu-west-3 (Paris, France)** or **eu-central-1 (Frankfurt, Germany)** datacenters.

This safeguards compliance by ensuring that location logs, GPS track points, and sensor telemetry never cross outside European borders for storage.



PostgreSQL Row-Level Security (RLS)

SECURE ISOLATION

Supabase's core PostgreSQL engine utilizes strict, hardware-level Row-Level Security rules. By binding security policies directly to database tables, the system guarantees physical isolation:

```
CREATE POLICY tenant_isolation ON traffic_logs
FOR SELECT USING (tenant_id = auth.uid());
```

Even if the frontend web server is compromised, Client A is physically blocked from querying Client B's logs, safeguarding proprietary sensor payloads.



Data at Rest

All database tables and backups are encrypted via industry-standard AES-256 algorithms.



Data in Transit

All client connections and API pushes are locked to encrypted TLS 1.2+ endpoints.



4. Project Roadmap & Financial Terms

Phased development outline and milestone billing structures

The total development cost for this build is **INR 400,000** (approx. €4,450). Payments are structured in four distinct milestones to guarantee continuous project alignment and testing validation:

- Milestone 1: Project Kickoff** 25% Advance
Initiates architectural mapping, database schema design, and server-side resource provisioning. Setting up authentication routes.
📺 Payment: ₹100,000 📅 Week 1-2
- Milestone 2: High-Fidelity Prototype** 25% at Delivery
Delivery of a live, unauthenticated HTML/CSS/Leaflet mockup on a staging link to finalize dashboard UX, Leaflet map interactions, and responsive mobile layouts.
📺 Payment: ₹100,000 📅 Week 3-4
- Milestone 3: Alpha Test & Core Engineering** 25% at Delivery
Integration of Supabase databases, Next.js routes, and Cloudflare Pages. Delivery of a functional staging portal where test files can be securely uploaded, parsed, mapped, and filtered. Data-sync API finalized.
📺 Payment: ₹100,000 📅 Week 5-7
- Milestone 4: Handover & Launch** 25% at Completion
Implementation of the Progressive Web App (PWA) installation manifest, Cloudflare WAF threat rules, final regression bug patching, and system migration to the live domain.
📺 Payment: ₹100,000 📅 Week 8-9



5. Infrastructure Limits & Managed SLA

Post-delivery support timelines, maintenance, and server limits

The setup leverages generous Cloudflare and Supabase Free Tiers, sufficient for standard business traffic. Under this setup, the platform supports the limits shown below.



Free Tier Infrastructure Capabilities

OPTIMIZED

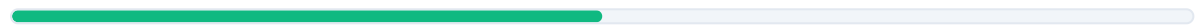
Bandwidth (Data Egress)

Up to 5 GB / month (10,000 requests)



Database Relational Storage

500 MB (Highly Compressed Postgres Tables)



Monthly Active Authenticated Users

Up to 50,000 MAUs



Compute Edge Functions

100,000 execution requests / day



Warranty Period

The first **3 months** post-handover are fully covered at **no additional cost**. This includes software bugs, deployment stabilization, and runtime debugging.

Ongoing Maintenance

Transition to a Managed SLA at **€100/mo** (or **€1,000/year**). Includes dependency security updates, database back-ups, and uptime audits.



Overage & Scaling Policy

Should logs or user volume breach free limits, upgrade to the Supabase Pro Tier is required (approx. €25/month + usage). We will provide a 30-day monitoring warning before limit thresholds are reached.